

How are library cataloging metadata used differently over time: a large-scale quantitative analysis of the Library of Congress catalog

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Abstract

[Purpose] The library catalog is not only a useful tool for patrons to find and access library collections but also a valuable dataset for various types of quantitative analysis on our cultural assets. This research aims to develop the latter potential of library metadata by conducting a large-scale analysis of how the metadata fields encoded in the Machine-Readable Cataloging (MARC) standards.

[Design/methodology/approach] We examined more than 6 million book records from 1980 to 2018 in the catalog of the Library of Congress. We particularly focused on how bibliographic records have changed over time and whether these changes correlate with the introduction of new cataloging policies.

[Findings] Our results show that more than 200 unique fields and 1,300 unique subfields have been used in the MARC format, though the majority of them are used in fewer than 1% of all records. At the same time, bibliographic records have become increasingly complex, with more fields and subfields per record over the past 40 years. Additionally, there are clear, although sometimes asynchronous, parallel developments between MARC tags and cataloging standards.

[Originality/value] This study represents the first large-scale quantitative analysis of the history of library metadata, revealing significant and interesting changes over time and highlighting challenges for the meaningful use of library metadata in the current data environment.

Keywords: Library cataloging, metadata, data science, MARC

1. Introduction

Libraries have been one of the most important repositories of information resources and cultural assets in human society. Despite the fast-changing landscape of information technologies and access, collections in libraries remain critical not only for their immediate users but also for society at large (Brophy & Bawden, 2005; Oakleaf, 2010). Vital to the use of these collections is the accompanying metadata, “structured information that describes, explains, locates or otherwise makes it easier to retrieve, use, or manage information resources” (Riley, 2004, p. 1). Library metadata typically takes the form of bibliographic records and is created and maintained through cataloging work undertaken by information professionals (Milstead & Feldman, 1999). Being a central part of the technical services in libraries, cataloging holds a particularly important role in meeting the needs of library patrons (Hoffman, 2009), powering the library catalogs that patrons use to understand and gain access to the library’s collection of resources.

Given the long history of professional practices creating library catalogs and other information discovery tools (Strout, 1956), it is not surprising that there has also been a long history of changing cataloging standards. For Anglo-American countries, the Anglo-American Cataloging Rules, 2nd edition (AACR2), first published in 1978, is heavily regarded as a major milestone, covering many resource types in the library collection and offering firm guidance for the cataloging community (Huthwaite, 2003; Tillett, 2003). In response to the growing concerns that library metadata practice was inconsistent with Internet technologies, a new cataloging standard, Resource Description and Access (RDA), was developed and eventually implemented in 2013. RDA is believed to be a critical link between the library cataloging community and the dominant linked data paradigm on the Internet (Tillett, 2013).

Juxtaposed with these recent, significant changes in cataloging standards is the persistence of a long-used encoding standard to render bibliographic records in electronic form and allow them to be exchanged between libraries (Aalberg & Žumer, 2013). The MACHine-Readable Cataloging (MARC) standard has been used as the primary encoding standard for library data since its introduction in the 1960s (Avram, 1968; Seikel & Steele, 2011). This continuity of format has provided consistency and interoperability for the library community and their data, but at the same time has driven bibliographic records further away from more modern data formats and practices (Tennant, 2002). In recent years, the Library of Congress (LC) has been actively involved in the development of BIBFRAME, a new approach to encoding bibliographic data more in line with Semantic Web approaches to data publication (Kroeger, 2013). The implementation of such a standard would fundamentally shift the landscape for bibliographic data use and reuse online.

We should also stress that there are parallel cataloging traditions that are similar to the Anglo-American tradition, in terms of higher-level practice and historical trend, despite the very different cataloging standards being used. For example, the Universal MARC format (UNIMARC) is an important MARC format used beyond the Anglo-American countries (Kokabi, 1995) and its relationship with FRBR and linked data has also been discussed in recent years

(Dunsire & Willer, 2011; Le Pape, 2011). Beyond encoding standards, the International Standard Bibliographic Description (ISBD) is the basis of many national content standards for cataloging (Howarth, 2014). These standards represent a distinct cataloging traditions compared to those used in the US. Given the situation of our study in the American Library of Congress, however, our discussion will be focused on the cataloging developments and standards in the Anglo-American tradition.

Changes in cataloging standards and practice hold implications for understanding both the current state of bibliographic data and how it has developed over time. However, how exactly such changes are reflected in the real-world library metadata is far from known. This gap is particularly critical for the following two reasons. First, understanding the quality and history of this metadata is instrumental to transforming library metadata into findable, accessible, interoperable, and reusable (FAIR) datasets and using such data in any future data science services (Greenberg, 2017). Second, the library cataloging community is also moving into the linked data paradigm guided by the Functional Requirements of Bibliographic Records model (International Federation of Library Associations et al., 1998) and later the IFLA Library Reference Model (Riva et al., 2017). This major movement also requires a granular understanding of the library cataloging practice to move the transfer the bibliographic data into a distinct data structures.

This study, part of a broader ongoing project aiming to develop large bibliographic collections into research datasets for data science research, is intended to provide a better understanding of this data and how it has changed over time. This topic has not received much attention in the cataloging community with the exception of a study conducted by Mayernik (2011), who used 1,500 randomly selected MARC records to investigate the power-law distribution of MARC fields. In this research, we aim to significantly expand the research framework in terms of both the sample size and publication window. Specifically, we use more than 6 million book metadata records from the catalog of the Library of Congress, created from 1980 to 2018 to address our research questions. We believe this research will offer a much more comprehensive view of how MARC as well as cataloging standards have shifted during the past four decades. Specifically, we strive to answer the following three research questions:

RQ1: What major trends are observed in the use of MARC metadata fields and subfields in the bibliographic records for books from the Library of Congress catalog?

RQ2: How do changes in the use of major fields and subfields relate to changes in cataloging standards and policies?

RQ3: How is the use of fields correlated with provenance of MARC records?

This research aims to offer the first large-scale analysis of how metadata fields are used in the library book metadata information. Bibliographic data is a large and valuable dataset, though its research usage has typically been limited to those with deep domain knowledge. In explaining this data and better understanding its quality, we hope our findings will support more effective

and responsible reuse of bibliographic data by a broader community. In addition, these findings stand to shed fresh light on the history of cataloging standards and practices, which, in return, will contribute to future developments in cataloging policy.

2. Literature review

The effort to organize information resources has a long history, and can be traced as far back as the Sumerian civilization, where lists of books were compiled (Strout, 1956). Efforts to organize information remain, and in modern times libraries have been constructed not only for preserving information assets, but also for providing access to such resources to broader society.

Cataloging standards have a long and varied history depending on the institution and collection requirements. The earliest modern library catalog in Western countries can be traced to Thomas Hyde's catalog for the Bodleian Library at Oxford in the 17th century (Smiraglia, 2003). In the modern era, the cataloging standard AACR was released in 1967 as a joint publication of the American, British, and Canadian library associations to further standardize cataloging practice in Anglo-American libraries (American Library Association et al., 1967). This was followed by AACR2, which is commonly regarded as a major milestone in cataloging as it covered many resource types in the library collection and offered strong support for the practices and traditions of the cataloging community (Huthwaite, 2003; Tillett, 2003).

Complimenting modern cataloging standards has been the encoding standard Machine-Readable Cataloging (MARC). MARC was developed in the 1960s by LC to enable electronic catalog records and automate the production of library catalogs (McCallum, 2010). The MARC standard was then independently developed in various countries, such as USMARC and UKMARC developed in the US and UK, respectively (Campos et al., 1995). It only became a consistent, international standard, now known as MARC21, at the end of the 20th century (Seikel & Steele, 2011). This standard remains the de facto encoding standard of libraries to this day and continues to undergo regular updates and revisions. Given the longevity of the MARC standard, it has left a strong mark on the practice of library cataloging. Specifically, MARC21 has remained the prevalent format long after its parallel development with AACR2, despite criticisms of its limitations and especially anachronism in the current technological environment (Tennant, 2002).

During the late 1990s, AACR2 was reevaluated to address new challenges associated with emerging digital resources and environments. Specifically, it was argued that the stagnation of the development of library cataloging standards in relation to internet technologies had made library catalogs a data silo, one that cannot be easily connected to the rest of the metadata environment (Rasmussen Pennington & Cagnazzo, 2019). The development of the new cataloging standard to replace AACR2 was heavily informed by the Functional Requirements for Bibliographic Records (FRBR) model, which is rooted in the user-centric philosophy and considers the current data structure (Dunsire, 2014). The outcome of this process was the cataloging standard Resource Description and Access (RDA), which was eventually

implemented in 2013. RDA is believed to be a critical link between the library cataloging community and the dominant linked data paradigm on the Internet (Tillett, 2013). In addition, a new encoding standard to replace MARC, Bibliographic Framework (BIBFRAME), has also been developed and heavily discussed during the past decade (Kim et al., 2021; Kroeger, 2013).

As the library cataloging community is moving towards a new paradigm exemplified by RDA and BIBFRAME, we argue that it is equally critical to reconsider library metadata from the perspective of data science. Data science is a heavily interdisciplinary research space that is made possible by the growing amount of data and focuses on using new technologies (particularly quantitative methods) to support the full lifecycle of large-scale datasets (Cao, 2017). This perspective is particularly relevant as the whole library and information science field becomes more heavily integrated with data science (Virkus & Garoufallou, 2020; Wang, 2018). In addition, the richness and quality of metadata available from library catalogs can make it a valuable addition to various data science applications (Qin & D'ignazio, 2010).

We are particularly inspired by the work of Mayernik (2011) to examine the distribution of MARC fields in bibliographic records. The author analyzed 1,500 MARC records and found a power-law distribution of the fields: very few fields available in the MARC format are used very frequently whereas most of the fields are used in only a very small percentage of records. This is the most recent large-scale analysis of MARC data, though it predates the switch to RDA cataloging rules and the accompanying changes in the MARC standard. In the present research, we aim to systematically expand this prior effort by looking at the entire LC catalog and analyzing how the cataloging practice shifts over a four-decade window, i.e., from 1980 to 2018. This time period reflects not only the change from AACR2 to RDA cataloging rules, but numerous changes to the MARC format as well.

3. Methods

This research is based on the library catalog of the Library of Congress. We chose LC because (1) it is one of “the largest and most international libraries in the world” (Cole, 2005) and as such this dataset is a good representation of the bibliographic universe, and (2) LC is one of the key players in the development and management of modern cataloging standards, such as AACR2, RDA, and MARC, and we expect that the bibliographic records in its catalog accurately reflect these cataloging standards, and (3) the library supplies its catalog as an easily accessible, open-access dataset. Given the large number of records made publicly available (over 25 million), researchers chose to limit the present examination to just book records, leaving analysis of additional formats for future consideration. Books present the best starting format for this analysis because they represent the highest proportion of records, and at the same time, it is the most classic type of resource in the library catalog and one that has been best supported by library cataloging standards (Frost & Taylor, 1983; Smiraglia, 2009).

We used the 2019 version of the MDSCONnect book metadata, the open-access provision of the MARC Distribution Services (MDS) supplied by the Library of Congress (Library of Congress

Cataloging Distribution Service, 2019). The raw dataset contains 11.28 million full bibliographic records in the MARC format from the Library of Congress catalog. While the Library of Congress offers several different file formats, we selected the MARCXML version. For each record, we parsed the MARCXML and counted the number of fields and subfields. We also selected key fields and subfields for a more dedicated examination.

From this dataset, we selected all book records that (1) were published between 1980 to 2018 and (2) had an *encoding level* (the 17th digit of the leader) of either full (value of empty space) or core (value of 4). By applying these two criteria, we sought to compile a more recent and well curated sample. Specifically, the second criterion limits our sample to those records that are considered complete and have been fully examined by at least one professional cataloger. Our final sample contained 6,475,931 records. Figure 1 shows the temporal distribution of our final sample (Panel A) as well as the share of these records within the raw dataset that were selected for the sample (Panel B). Our selection criteria create a bias against the most recent publications, which have not had as much time to receive fully vetted, high-quality metadata records, as indicated by their encoding level. In particular, significantly fewer books published after 2010 are represented in the dataset, especially in 2018, which may result in inconsistent patterns from the analysis.

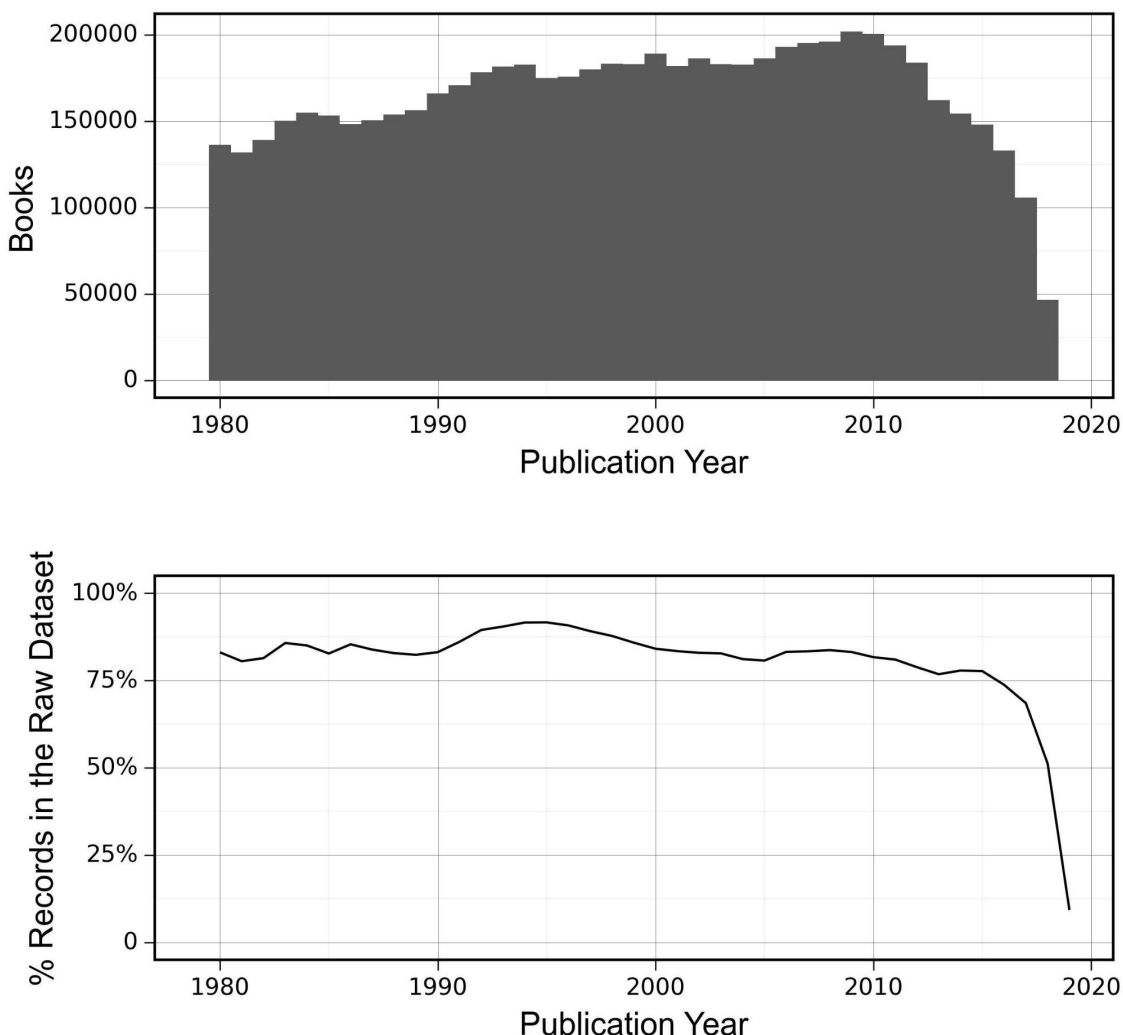


Figure 1: Temporal distribution of our sample in terms of number of books (top) and the share of the sample in the raw dataset (below). Source: Authors own work

In our analysis of the contents of these records, we were guided by the literature on changes to cataloging standards within the sampled time period. This led researchers to focus specific attention on not only the most common fields, but also key fields and subfields within MARC that have been heavily impacted during the past few decades. Examples include the 264 field as the replacement of the 260 field under RDA, the new 336-338 fields being introduced with RDA, and the 440 field being replaced by the 490 field in the 2000s. Such cases offer insight into how library metadata in practice has shifted under the changing cataloging standards.

We further constructed a linear regression model to understand the statistical relationship between the use of fields (as dependent variable) and the history and provenance of records. In particular, we use the following three independent variables: (1) publication year of the book (taken from 008 field of MARC record), (2) the time difference between the file date (taken from 008 field) and the last transaction date (005 field) in the unit of year, and (3) the number of editing agencies (040-d). The latter two variables offer a comprehensive recording of the

provenance of the cataloging record, especially how many edits a record received after its creation. All data analysis and visualization were performed using Jupyter Notebook.

4. Results

4.1 The overview of how fields and subfields are used

Our final sample contained 6.48 million book records; these records contained a total of 112.2 million fields and 225.6 million subfields. On average, each record contained 17.3 and 34.8 fields and subfields respectively. As shown in Figure 2, the highest density of records occurs at 15 fields, and at 26 subfields. It should be noted that there is a long tail in these distributions: the largest number of fields in a single record is 194, the majority of which are parallel 020 fields (ISBNs). On the other side of the spectrum, the smallest number of fields in a record observed was 4, which implies a less than full level of cataloging and is likely an error in the data.

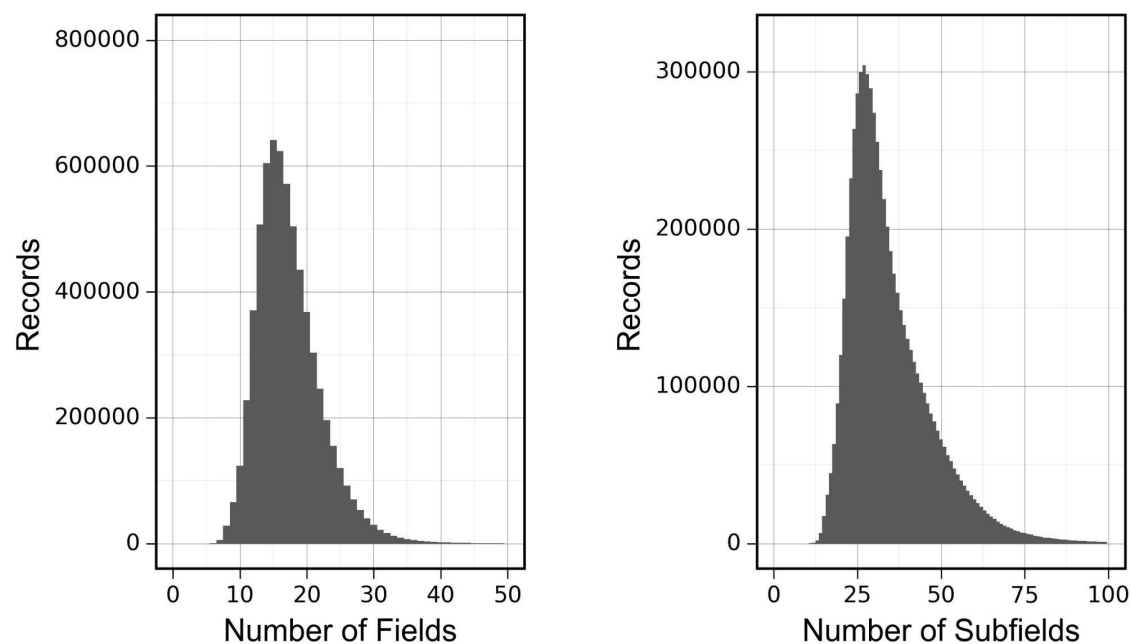


Figure 2: Distribution of records. Source: Authors own work

We further examined the pattern across the top 10 book languages (as in 008 field) identified from the database. We found that Chinese, Japanese, and Korean language publications have significantly more fields used than the other languages. A closer investigation shows that the difference can be largely explained by the 880 field (Alternate Graphic Representation) required for publications in a different script. After removing the 880 field, all top 10 languages have very similar number of fields, as shown in Figure 3.

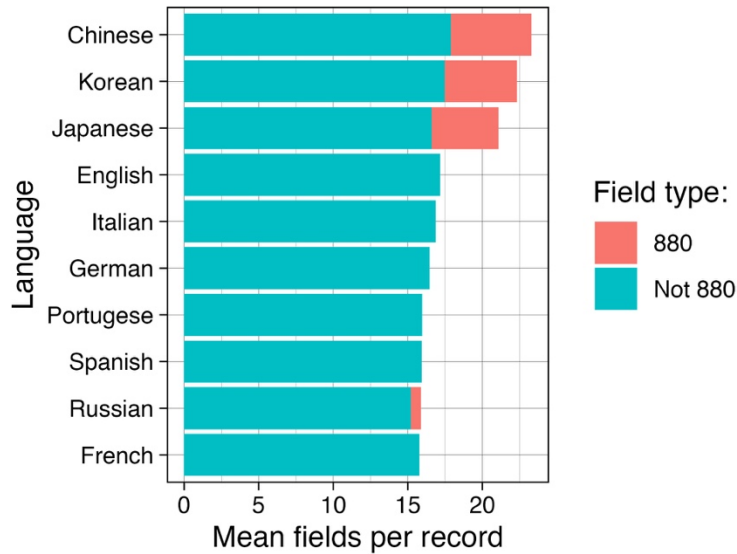


Figure 3: Use of fields by publication language. Source: Authors own work

Among all records, we found 203 unique fields and 1,337 unique subfields within our sample, showing the breadth and diversity of information available in the library metadata. Backing up Mayernik’s (2011) earlier findings, certain fields and subfields were far more likely to occur than others. In fact, within all records in the sample, fields and subfields are used in a highly uneven manner: only 5.9% of all fields and 1.6% of all subfields are used in more than 50% of records. Tables 1 and 2 show the top 10 most commonly used fields and subfields within the sample.

Table 1: Top 10 fields used in the collection. Source: Authors own work

Fields	Field Name	Records	Share of records
010	Library of Congress Control Number (LCCN)	6,475,931	100.0%
245	Title Statement	6,475,931	100.0%
050	Library of Congress Call Number	6,474,192	100.0%
300	Physical Description	6,473,886	100.0%
040	Cataloging Source	6,081,946	93.9%
260	Publication, Distribution, etc. (Imprint)	5,564,780	85.9%
020	International Standard Book Number	5,384,287	83.1%
650	Subject Added Entry-Topical Term	5,321,388	82.2%

100	Main Entry - Personal Name	4,613,081	71.2%
504	Bibliography, etc. Note	3,934,461	60.8%

Table 2: Top 10 subfields used in the collection. Source: Authors own work

Subfield	Subfield Name	Records	Share of records
010-a	LC control number	6,475,931	100.0%
245-a	Title	6,475,931	100.0%
050-a	Classification number	6,474,192	100.0%
300-a	Extent	6,473,871	100.0%
300-c	Dimensions	6,448,053	99.6%
050-b	Item number	6,437,742	99.4%
040-a	Original cataloging agency	6,080,315	93.9%
040-c	Transcribing agency	6,080,028	93.9%
245-c	Statement of responsibility, etc.	6,039,984	93.3%
260-a	Place of publication, distribution, etc.	5,563,575	85.9%

While a small number of fields and subfields are common across the sample, many others occur quite rarely: 85.2% of 95.4% of all fields and subfields, respectively, are used in fewer than 10% of all records. Tables 3 and 4 illustrate the number of fields and subfields used above the thresholds of 90%, 50%, 10%, and 1% of all records. Of particular note is that a large proportion of subfields occur in less than 1% of all records. This shows that many metadata elements included in the library cataloging standards are highly specific to special situations, even in the book format, again echoing findings from Mayernik (2011).

Table 3: Summary of the use of fields

Threshold	Fields	Share
Over 90% records	7	3.4%
Over 50% records	14	6.9%
Over 10% records	30	14.8%

Over 1% records	53	26.1%
All fields	203	100%

Table 4: Summary of the use of subfields. Source: Authors own work

Threshold	Subfields	Share
Over 90% records	15	1.1%
Over 50% records	28	2.1%
Over 10% records	65	4.9%
Over 1% records	150	11.2%
All subfields	1,337	100%

Finally, it should be noted that not every field or subfield observed is current or correct. We found subfields that are obviously wrong based on the MARC format, such as 019 subfield a (019-a) which has never been in use at LC. These subfields appear in very few records, but demonstrate that some errors exist in this dataset, and that even LC's robust quality control practices cannot prevent all MARC errors.

4.2 Use of metadata over time

There is an increase in the amount of metadata elements in MARC records over time, as shown in Figure 3. Both the numbers of fields and subfields per record increased by around 100% during the past four decades. In 1980, the average number of fields per record was 13, but by 2018 this number had risen to over 20. Similarly, the number of subfields per record doubled during this time. Some of this increase may be attributed to increased repetition of certain fields. In 1980, any given field appeared an average of 1.12 times per record, though this number rises to 1.22 in 2018. This shows that records now contain more repeated fields than before.

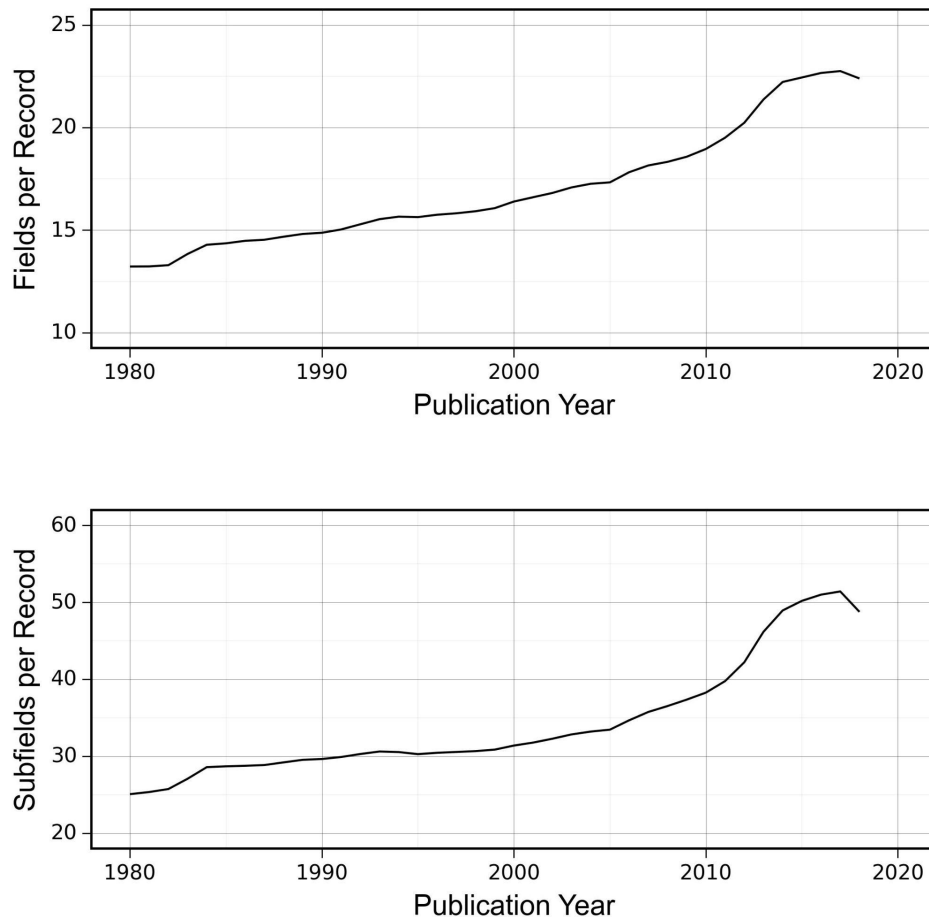


Figure 3: Number of fields and subfields per record (excluding fixed-length fields).
Source: Authors own work

To provide a closer look at trends most likely to impact more records, we next focused on how the ten most frequently used fields (see Table 1) are used over the time period examined, with the results shown below in Figure 4. While some of the fields are used very consistently over time, such as 010 field for the Library of Congress Control Number and 245 field for the title and the statement of responsibility, there are some significant changes in the data as well. One notable example is the field 260 (publication statement), which was replaced by the similarly functioning 264 field due in part to the change from AACR2 to RDA in the early 2010s; this will be further explored below in the Discussion.

Some other changes can be explained by the shift of cataloging policies and practice as well, such as the increasing use of ISBN numbers (field 020) in the global book industry (Ehlers, 2000) and the increasing requirement for subject information in cataloging standards, as encoded in the 650 and neighboring fields. Some patterns might be explained by changing data quality control practices at LC, for example, the low usage of the required 040 field during the 1980s that was resolved by 1990.

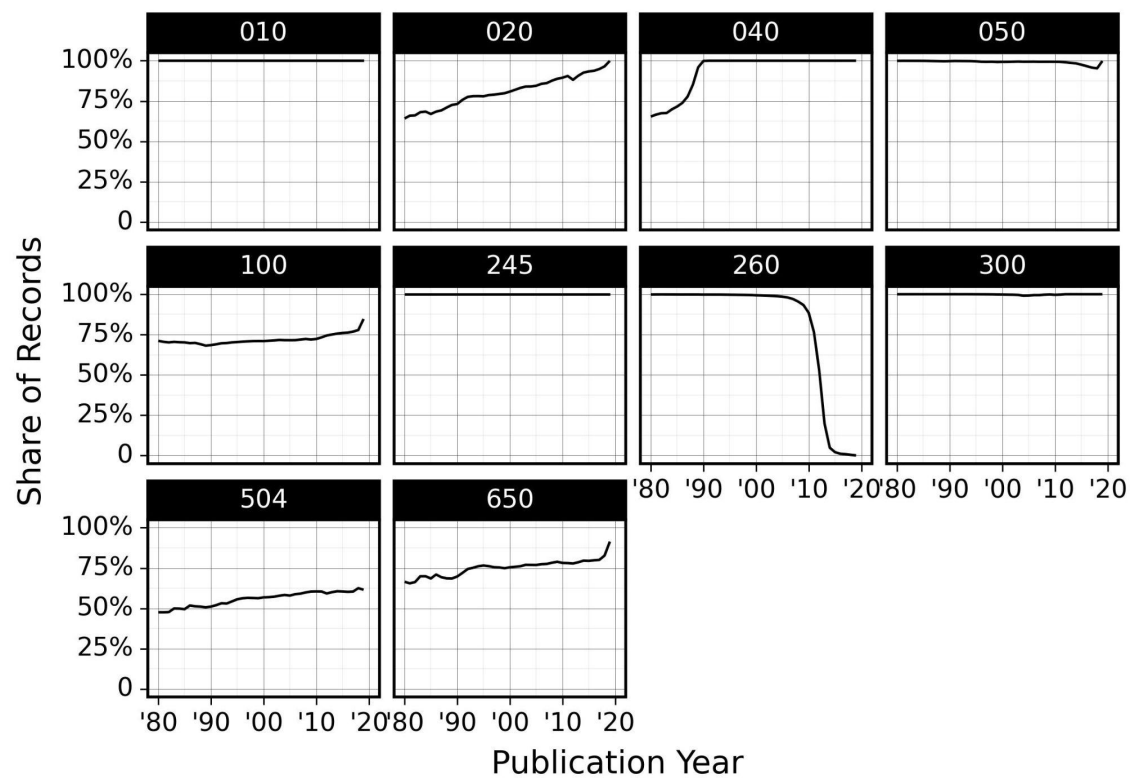


Figure 4: Use of top 10 fields over time. Source: Authors own work

Figure 5 shows the usage of the 10 most common subfields over time.

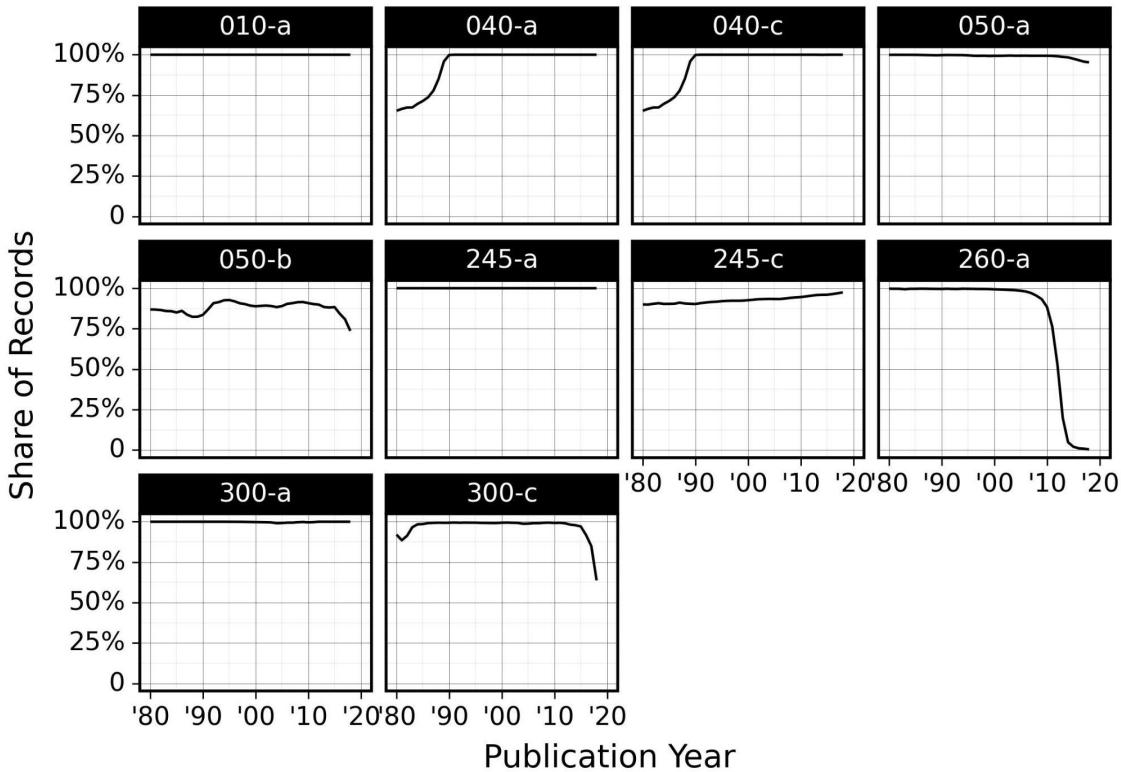


Figure 5: Use of top 10 subfields over time. Source: Authors own work

While the drop in usage of subfield 260-a observed here is explained by the aforementioned shift away from the use of this field entirely, other changes, such as the drop in usage of 050-b (LCC item number) and 300-c (physical measurements) may be explained by data quality control processes for the most recent records in the dataset; this will also be explored further in the Discussion.

To explore the statistical relationship between the use of fields and the history and provenance of MARC records, we developed a linear regression model based on the variables outlined in the Methods section. Our goal was to determine whether the number of fields used in a record changes over time and as the record receives more attention through editing. To mitigate the impact of sample size and imbalance, we applied a stratified sampling approach, randomly selecting 100 records from each year between 1980 and 2018. The final dataset consists of 3,900 records, which were used for analysis. The results are presented in Table 5.

Table 5: Results of linear regression model. Source: Authors own work

Variable	Estimate	95% Confidence	p-value
Intercept	-411.5	[-437.9, -385.1]	<0.001

Publication Year	0.213	[0.200, 0.227]	<0.001
040-d agencies	1.055	[0.981, 1.129]	<0.001
Date difference (in year)	0.075	[0.058, 0.092]	<0.001

The 'Estimate' column indicates how many additional fields are used in a record when each of the three independent variables increases by one unit. The '95% Confidence Interval' column provides the range of values within a 95% confidence level. The results show that all three independent variables have a statistically-significant positive correlation with the number of fields, confirming our earlier observation that MARC records use more fields over time. Additionally, the findings suggest that as a record undergoes more editing, it is likely to have more fields. This relationship holds true even when (1) using the entire LC catalog population instead of the selected sample and (2) treating the publication year as a categorical variable. Both of the latter checks reinforce the validity of the above results.

4.3 Examination of key fields

Next, we examined fields associated with some key changes in the MARC format during the past few decades. Figure 6 shows the use of 336, 337, and 338 fields, which offer granular details of a resource's format and were first introduced as required metadata fields by RDA (Chapman et al., 2006). With the implementation of RDA in US libraries in January, 2013 ("U.S. RDA Implementation Updates from the U.S. RDA Test Coordinating Committee.," 2016), we can observe the strong presence of these fields in our dataset from the beginning of the 2010s; they are present in virtually every record by 2015.

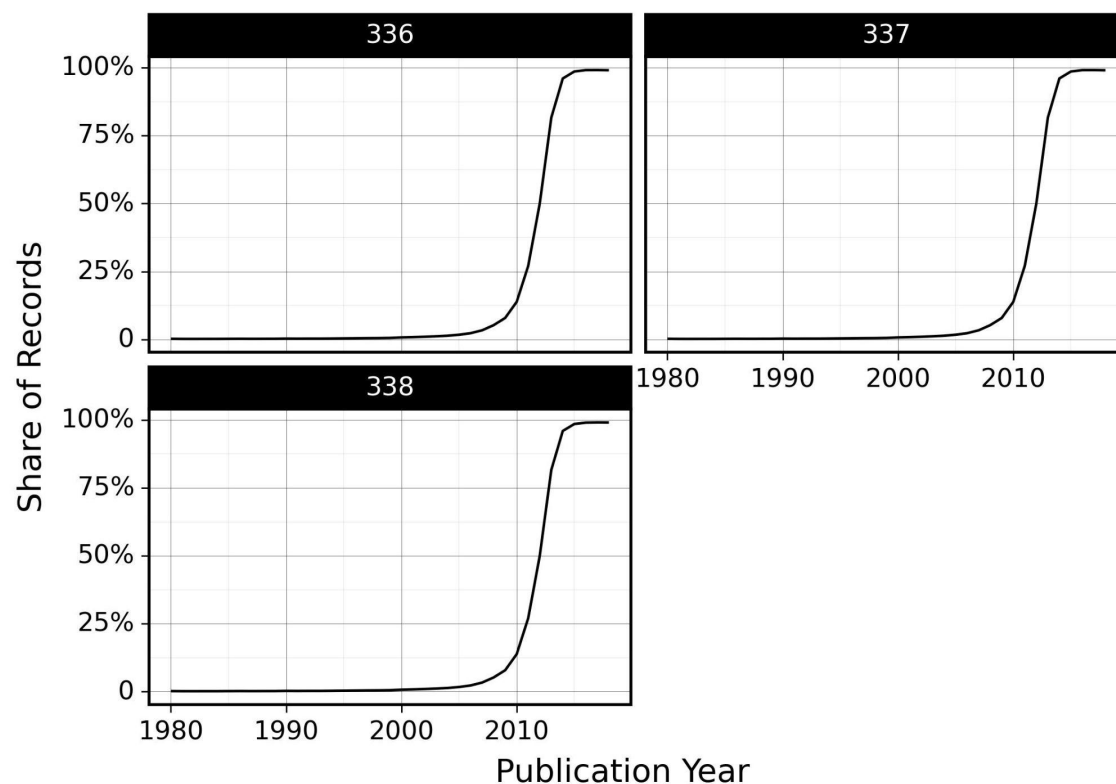


Figure 6: Share of records with 336-338 fields. Source: Authors own work

Field 264 was introduced into the MARC format to record various types of publication and production information (Library of Congress, 2024), a function that was previously performed using field 260. Similar to the 33X fields discussed above, we observe a strong shift in the use of these fields in the beginning of the 2010s (Figure 7), corresponding again to the implementation of RDA in U.S. libraries.

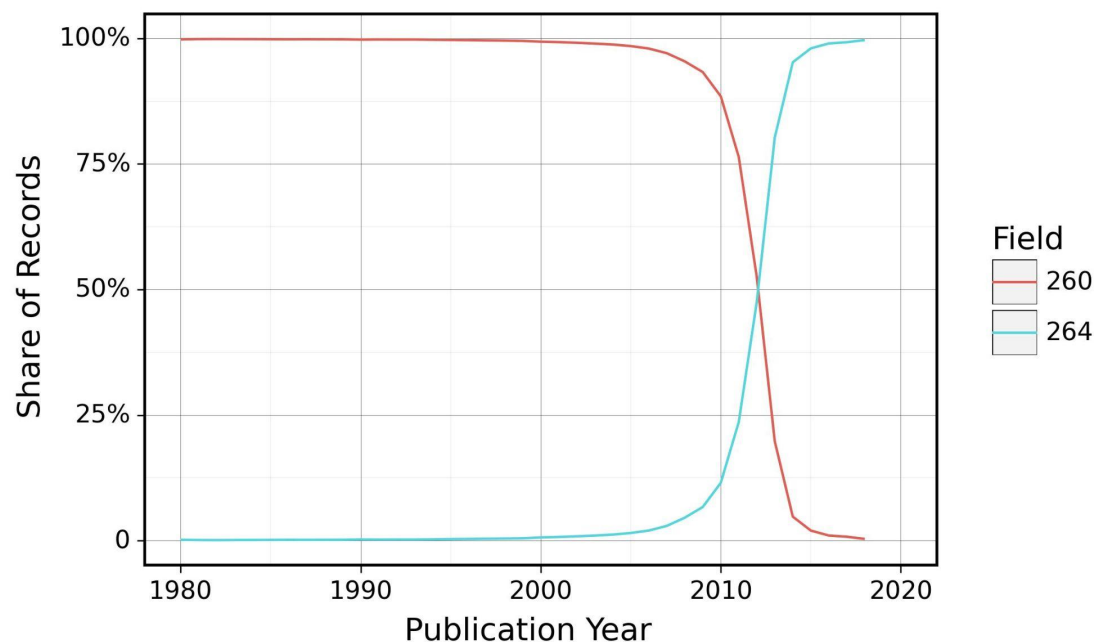


Figure 7: Share of records with 260 and 264 fields. Source: Authors own work

Another expected shift in field usage concerns series data, which has been associated with a number of different fields over time. In 2008, the 440 field (series name/added entry) was rendered obsolete, resulting in greater use of other series fields to record this information, including 490 (series statement) and 830 (controlled series added entry) (Library of Congress, 2024). Figure 8 shows the changes in usage among these three fields over time.

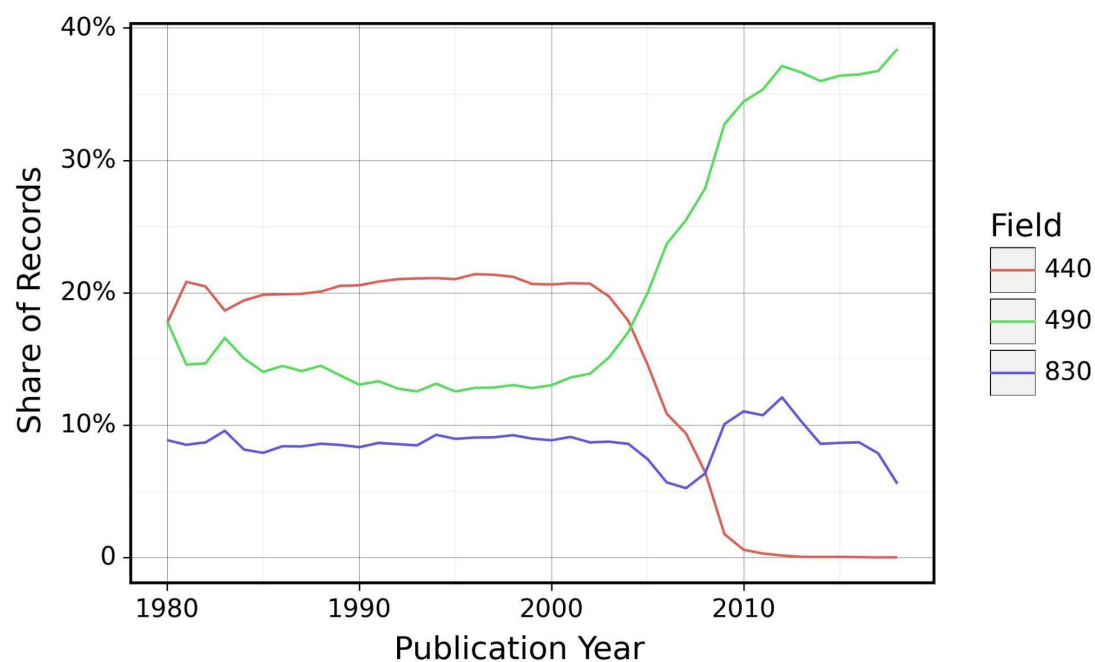


Figure 8: Share of records with 440 and 490 fields. Source: Authors own work

Recent years have seen an increased emphasis on subject and genre metadata for all materials, including fictional works (DeZelar-Tiedman, 2011). In particular, genre has received more attention in cataloging since the consolidation of a number of genre thesauri into the Library of Congress Genre/Form Terms (LCGFT) starting in 2007 (Bitter & Tosaka, 2020). Figure 9 depicts two trends related specifically to the 655 field used to encode genre data. First, the proportion of records with at least one 655 has grown from less than 5% to close to 20% since 1980. Second, the proportion of all subject fields (6XX) that are 655 has similarly grown, with genre fields now making up between 10% and 15% of all subject metadata. Overall, the graph shows that more genre information is being included in bibliographic records, especially since 2010.

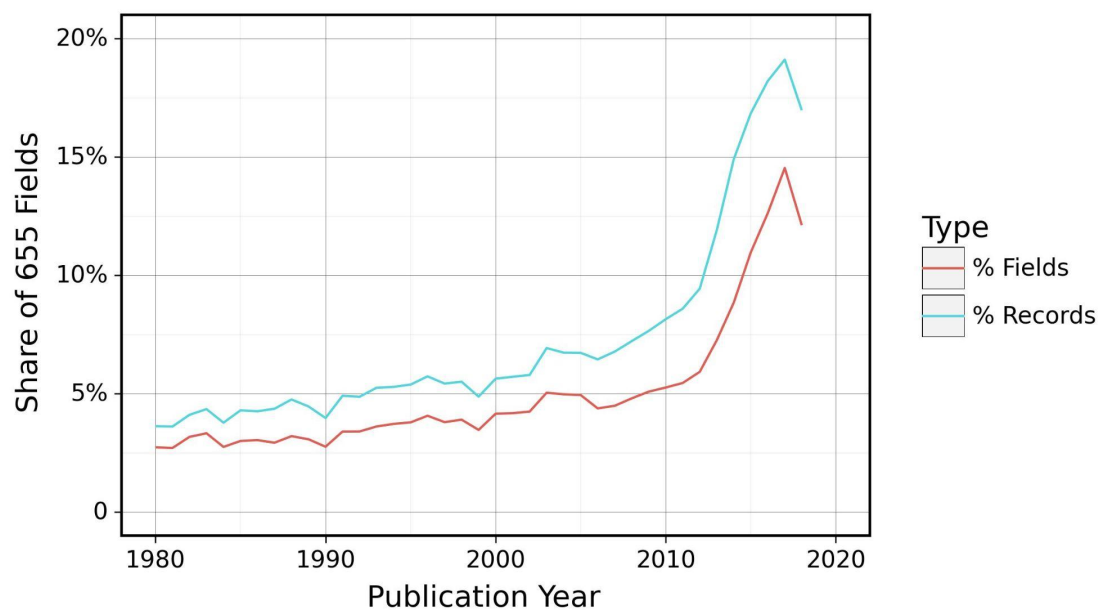


Figure 9: Use of 655 fields over time. Source: Authors own work

In bibliographic records, the 7XX fields (including 700, 710, 711, and 730), are used to record additional access points (beyond the primary access point) for related names and titles, and their values are typically placed in indexes used by Title and Author searches in catalogs. Unlike many previously observed field trends, the 7XX fields show a pattern of declining usage in recent years. Figure 10 shows two trends related to the 7XX fields. First, after growing between 1980 and 1990, the number of records with at least one 7XX field has declined, particularly since 2010. Second, as shown by the “% Fields” line, the proportion of 7XX fields to all access point fields has declined. This means bibliographic records now have, on average, fewer additional name and title access points. The implications of this finding are discussed in the following section.

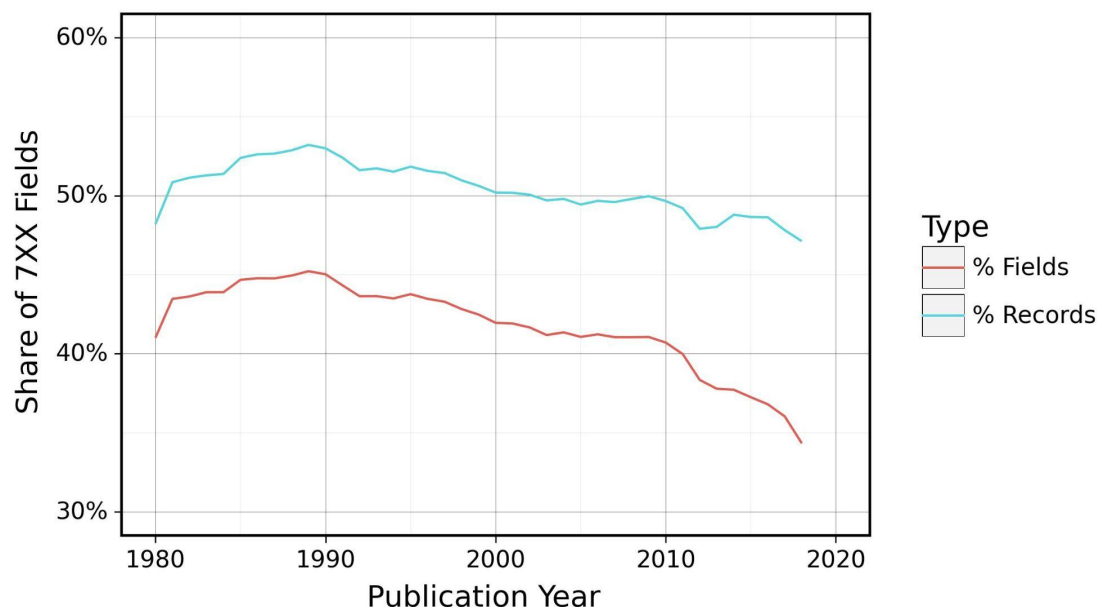


Figure 10: Use of 7XX fields over time. Source: Authors own work

5. Discussion

MARC is a robust and long-lived standard, capable of encoding complex bibliographic metadata in an array of granular fields and subfields. While 203 unique fields and 1,337 unique subfields were observed across LC's book dataset, very few of them were observed with any regularity. This general pattern of usage confirms findings from previous work in this area (Mayernik, 2011). The majority of fields and subfield instances observed belong to a small subset designed to encode some of the most basic bibliographic data such as title, author, publisher, physical description, and call number. This is true across languages, when fields specifically devoted to alternate script representations are accounted for. Even so, the large number of different subfields used across the dataset demonstrates the expressiveness of the MARC encoding format. Many metadata elements included in library cataloging are highly specific to special situations, underscoring both the deep format knowledge needed by catalogers and the specialized nature of bibliographic data. At the same time, this broad pattern of field usage may pose some problems for the reuse of this data. Mayernik (2011) argued that the low usage of some MARC fields (such as 130, 240, and 76X-78X) make it difficult to translate the MARC format into FRBR, where such information is essential to illustrate the relationship between entities. In addition, the very sparse nature of library metadata (i.e., many null values in metadata fields) makes it difficult for it to be transformed into databases and used in data analysis and machine learning algorithms (Beckmann, 2006; Greenland et al., 2016).

A number of usage patterns demonstrate changes in how the MARC standard has been used, particularly in the past decade. The number of fields and subfields in a typical record has grown over time, especially since the early 2000s. This suggests bibliographic records now contain more granularly encoded data than before. Some of this trend can be attributed to the new fields

and subfields introduced during transition to the RDA cataloging rules. For example, the 33X fields (see Figure 6) are a set of fields first introduced in 2009 and designed to encode specific metadata on content and format prescribed by RDA. As these elements of metadata are either required or strongly recommended in all RDA descriptions, this adds three additional MARC fields to most records created after the widespread adoption of RDA in 2013. An increased emphasis on genre data is another recent trend that may be driving an increase in the average number of fields and subfields per record. Since the introduction of LC's genre vocabulary LCGFT in 2007, genre metadata in bibliographic records has become more standardized and easier to include (Bitter & Tosaka, 2020). As shown above in Figure 9, more records now contain genre metadata that would have previously been omitted. Finally, consideration of record provenance shows that an individual bibliographic record tends to increase in length over time, especially as more cataloging agencies interact with that record.

Usage patterns observed during the present study also shed new light on the quality of bibliographic metadata. Library metadata, especially originating from LC, has a general reputation for the high levels of care and maintenance required in its creation, though conceptions of bibliographic record quality are somewhat more complicated (Snow, 2017). Patterns of field and subfield use in the LC dataset examined here show general consistency and adherence to requirements of the MARC format. While overall findings support the high quality nature of this dataset, some issues were observed during analysis. Several figures above (e.g., Figure 3, Figure 9) show a dip in field and subfield count in the latest years in the sample. A possible explanation is that the most recent records, while still full level cataloging, did not yet include all the metadata they would ultimately have. In some cases, though, this may be the result of LC quality control processes that had not been completed for the most recently created records. For example, researchers reviewed a sample of recent records shown to be lacking a 300-c subfield (measurements) (see Figure 5). Closer inspection revealed that this data was incorrectly encoded in either 300-a (number of pages) or 300-b (illustrations and other details). Incomplete cataloging or quality control processes may also explain dips observed in the usage of 050 subfields (call number data) and other fields and subfields. Such findings illustrate the complex nature of the MARC standard. It is difficult to prevent all errors and omissions, even for a well-resourced institution such as LC. Even bibliographic datasets among the highest quality, such as analyzed during this study, may exhibit some problems.

The past 15 years have seen a significant number of policy and practice changes regarding bibliographic records. Many of these changes are visible in the results of this study. A very clear example of the impact of RDA policy and practice on this dataset is observable in the fields used for encoding publication information. Prior to the adoption of RDA, the MARC 260 field was used to record publication and similar information. Under RDA, to more clearly differentiate between publisher and other roles such as distributor or manufacturer, a new field was created. The MARC 264 field, first introduced in 2011, would replace 260 and render it obsolete. The ensuing quick and total change between the two fields is illustrated well in Figure 7. The previously mentioned implementation of the 33X fields is yet another example of RDA's clear impact on the contents of this dataset. However, not all relevant changes in recent years are related to RDA. Other significant changes are related to updates to the MARC format and policy shifts at LC itself. For example, Figure 8 above shows the complicated interplay of three fields

used to record series information. Within the MARC format, 440 was rendered obsolete in 2008, driving series information to appear in 490 or 830 instead. However, around the same time, LC ceased creating and updating series authority records (Library of Congress, 2021). As a result, LC is more likely to record series information in an uncontrolled format (490) rather than an authority controlled format (830). For all of these reasons, we see a dramatic growth in 490 usage from around this time, and while 830 usage initially fell then rose, it has since declined again.

Still other patterns observed in the data show that important distinctions must be made between policy and actual practice. A good example of this is the previously mentioned genre element, field 655. No single policy change instructed catalogers to begin recording this information more. Rather, it is likely the increased awareness brought on by the creation of LCGFT, the ease of use of this new vocabulary, and the resulting greater discernment between subject and genre that has led to a greater attention to genre data in practice. While this may lead to increased access for users, other changes in practice may actually be reducing access. One of the benefits of RDA implementation was an expected increase in the amount and variety of access points (Oliver, 2010). As shown in Figure 10, however, the usage of 7XX name and title access point fields has actually decreased since RDA adoption in the early 2010s. While RDA may encourage the creation of more access points, in truth it renders many of them optional rather than required. As a result, catalogers may in practice be recording fewer name and title access points than they were under the previous cataloging rules, AACR2. Even where policy changes show up in expected ways in the dataset, users of bibliographic datasets must be aware that not all changes are retroactive. For example, while the 264 and 33X fields are present for all recent records, older records will not have these changes applied to them after the fact. For any large set of bibliographic records collected over time, users must be aware of when major policy and practice shifts occurred, as well as how they played out, to most effectively understand and use the data.

In addition to providing fresh insight into the history of MARC and cataloging practices, our work aims to demonstrate the potential of large-scale data analysis for the cataloging community. As previously discussed, the diverse and at times infrequently utilized metadata fields in library catalogs serve as a valuable resource for library patrons and their specific needs. The linked data movement (Möller et al., 2010) promises to make data usable not only to humans but also machine agents. However, the fact that many MARC fields and subfields have never been used in most of the records raises significant concerns about transforming MARC records into useful datasets in linked data environments. Considerations of this unique data format and its history must be taken into account moving forward. Library metadata, due to its high quality and richness, is a unique asset that the library community can contribute to the new revolution in information technologies. By treating the library catalog as a dataset and applying quantitative statistical and data analysis, we can create new value for both the library community (e.g., quality control and training new cataloging staff) and broader society (e.g., supporting empirical research in various knowledge domains and training new algorithms and services in cultural assets).

6. Conclusion

This paper presents a quantitative analysis of over 6 million MARC records from the Library of Congress catalog. By examining the usage patterns of MARC fields and subfield tags over time, we provide new and comprehensive evidence on the evolution of cataloging practices over the past four decades and their correlation with changes in cataloging policies. Our findings indicate a steady increase in the number of metadata fields and subfields utilized in library book records. Additionally, these records accurately reflect the evolving cataloging practices, despite the typically gradual nature of these changes.

To the best of our knowledge, this is the first large-scale quantitative analysis of library metadata. It offers a novel perspective on examining library metadata as a research dataset. We argue that this presents a promising opportunity for library metadata to create new relevance and connections to a broader community. As demonstrated in this paper, this approach can yield fresh insights into the nature of library metadata and pave the way for future possibilities with the aid of new information technologies. With linked data approaches and applications growing in prominence, bibliographic data represents a valuable source; better understanding the nature of this data can reduce the barriers to its reuse in the Semantic Web.

Despite the timely contribution of this paper, there are a few limitations to acknowledge. First, our analysis is confined to book metadata, a specific type of information resource in library catalogs. Since cataloging standards encompass a wide range of information resources, future research is warranted to understand how metadata for other types of library collections evolves over time and how it differs from book metadata. Second, our findings are based solely on the Library of Congress. While LC holds significant importance for the cataloging community, a broader comparison with other major libraries, particularly national libraries such as the British Library, could reveal more interesting findings about cataloging practices and data across international communities. This broader comparison is an important next step we aim to undertake.

References

- Aalberg, T., & Žumer, M. (2013). The value of MARC data, or, challenges of frbrisation. *Journal of Documentation*, 69(6), 851–872. <https://doi.org/10.1108/JD-05-2012-0053>
- American Library Association, Library of Congress, & Canadian Library Association. (1967). *Anglo-American cataloging rules*. Chicago, American Library Association.
- Avram, H. D. (1968). *The MARC II format : a communications format for bibliographic data*. Washington : Information Systems Office, Library of Congress.
- Beckmann, J. L. (2006). *Relational database management system support for sparse data sets*. <https://utk.idm.oclc.org/login?url=https://www.proquest.com/dissertations-theses/relational-database-management-system-support/docview/304984740/se-2>
- Bitter, C., & Tosaka, Y. (2020). Genre/Form Access in Library Catalogs: A Survey on the Current State of LCGFT Usage. *Library Resources & Technical Services*, 64(2), 44. <https://doi.org/10.5860/lrts.64n2.44>
- Brophy, J., & Bawden, D. (2005). Is Google enough? Comparison of an internet search engine

- with academic library resources. *Aslib Proceedings*, 57(6), 498–512.
<https://doi.org/10.1108/00012530510634235>
- Campos, F. M., Inês Lopes, M., & Galvão, R. M. (1995). MARC formats and their use: An overview. *Program*, 29(4), 445–459. <https://doi.org/10.1108/eb047213>
- Cao, L. (2017). Data science: a comprehensive overview. *ACM Computing Surveys (CSUR)*, 50(3), 1–42.
- Chapman, A., Kartus, E., Paradis, D., & Zinkham, H. (2006). *GMD/SMD Working Group: Proposal for Content and Carrier Terms in RDA*.
- Cole, J. Y. (2005). The Library of Congress Becomes a World Library, 1815-2005. *Libraries & Culture*, 40(3), 385–398. <https://doi.org/10.1353/lac.2005.0046>
- DeZelar-Tiedman, C. (2011). Subject access to fiction: An application of the Guidelines. *Library Resources & Technical Services*, 40(3), 203–210.
- Dunsire, G. (2014). *RDA and the semantic Web : lectio magistralis in library science*. Casalini Libri. <http://digital.casalini.it/9788876560132>
- Dunsire, G., & Willer, M. (2011). UNIMARC and linked data. *IFLA Journal*, 37(4), 314–326.
- Ehlers, H. J. (2000). Thirty years of ISBNs: The multiple parenthood of a success story. *Logos (London, England)*, 11(1), 25–27. <https://doi.org/10.2959/logo.2000.11.1.25>
- Frost, C. O., & Taylor, A. G. (1983). *Cataloging nonbook materials : problems in theory and practice*. Littleton, Colo. : Libraries Unlimited.
- Greenberg, J. (2017). Big Metadata, Smart Metadata, and Metadata Capital: Toward Greater Synergy Between Data Science and Metadata. *Journal of Data and Information Science*, 2(3), 19–36.
- Greenland, S., Mansournia, M. A., & Altman, D. G. (2016). Sparse data bias: a problem hiding in plain sight. *BMJ*, 352. <https://doi.org/10.1136/bmj.i1981>
- Hoffman, G. L. (2009). Meeting Users' Needs in Cataloging: What is the Right Thing to Do? *Cataloging & Classification Quarterly*, 47(7), 631–641.
<https://doi.org/10.1080/01639370903111999>
- Howarth, L. C. (2014). ISBD as bibliographic content standard: interweaving threads, contemplating a future. *Cataloging & Classification Quarterly*, 52(8), 982–999.
- Huthwaite, A. (2003). AACR2 and Other Metadata Standards: The Way Forward. *Cataloging & Classification Quarterly*, 36(3–4), 87–100. https://doi.org/10.1300/J104v36n03_08
- International Federation of Library Associations, Institutions Section on Cataloguing Standing Committee, & IFLA Study Group on the Functional Requirements for Bibliographic Records. (1998). *Functional requirements for bibliographic records: final report* (Vol. 19). KG Saur Verlag GmbH & Company.
- Kim, M., Chen, M., & Montgomery, D. (2021). Moving Toward BIBFRAME and a Linked Data Environment. In *Advances in library administration and organization* (Vol. 42, pp. 131–154). <https://doi.org/10.1108/S0732-067120210000042011>
- Kokabi, M. (1995). The internationalization of MARC: Part IV: UNIMARC, some formats based on it and some other MARC formats. *Library Review*, 44(7), 8–33.
- Kroeger, A. (2013). The Road to BIBFRAME: The Evolution of the Idea of Bibliographic Transition into a Post-MARC Future. *Cataloging & Classification Quarterly*, 51(8), 873–890. <https://doi.org/10.1080/01639374.2013.823584>
- Le Pape, P. (2011). Expressing FRBR in UNIMARC Yes we can! *IFLA WLIC 2011, San Juan, Puerto Rico*.
- Library of Congress. (2021). *Library of Congress Policy Regarding Requests for the Creation and Modification of Catalog Records*. <https://www.loc.gov/catdir/cpsol/modification-catalog-records.html>
- Library of Congress. (2024). *MARC 21 Format for Bibliographic Data*. Library of Congress. <https://www.loc.gov/marc/bibliographic/>
- Library of Congress Cataloging Distribution Service. (2019). *Books all : MDS Connect dataset*,

- MARC Distribution Services (data set) books all. Library of Congress. Cataloging Distribution Service. <https://www.loc.gov/item/2020445551>
- Mayernik, M. (2011). The distributions of MARC fields in bibliographic records. *Library Resources & Technical Services*, 54(1), 40–54.
- McCallum, S. (2010). Machine Readable Cataloging (MARC): 1975-2007. In M. J. Bates & M. N. Maack (Eds.), *In Encyclopedia of Library and Information Sciences* (Third, pp. 3530–3539). CRC Press.
- Milstead, J., & Feldman, S. (1999). Metadata: cataloging by any other name. . . . In *Online* (Vol. 23, Issue 1, pp. 24–26). Weston, Conn: Online, Inc.
- Möller, K., Hausenblas, M., Cyganiak, R., & Handschuh, S. (2010). Learning from linked open data usage: Patterns & metrics. *Web Science Conference 2010*, 1–8.
- Oakleaf, M. (2010). *The Value of Academic Libraries: A Comprehensive Research Review and Report*. https://www.ala.org/sites/default/files/2024-06/val_report.pdf
- Oliver, C. (2010). *Introducing RDA : a guide to the basics*. American Library Association.
- Qin, J., & D'ignazio, J. (2010). The Central Role of Metadata in a Science Data Literacy Course. *Journal of Library Metadata*, 10(2–3), 188–204. <https://doi.org/10.1080/19386389.2010.506379>
- Rasmussen Pennington, D., & Cagnazzo, L. (2019). Connecting the silos. *Journal of Documentation*, 75(3), 643–666. <https://doi.org/10.1108/JD-07-2018-0117>
- Riley, J. (2004). Understanding Metadata: What is Metadata, and What is it For? In *NISO Press*. National Information Standards Organization (NISO). [https://groups.niso.org/higherlogic/ws/public/download/17446/Understanding Metadata.pdf](https://groups.niso.org/higherlogic/ws/public/download/17446/Understanding%20Metadata.pdf)
- Riva, P., Le Boëuf, P., & Žumer, M. (2017). *IFLA Library Reference Model*.
- Seikel, M., & Steele, T. (2011). How MARC Has Changed: The History of the Format and Its Forthcoming Relationship to RDA. *Technical Services Quarterly*, 28(3), 322–334. <https://doi.org/10.1080/07317131.2011.574519>
- Smiraglia, R. P. (2003). The History of “The Work” in the Modern Catalog. *Cataloging & Classification Quarterly*, 35(3–4), 553–567. https://doi.org/10.1300/J104v35n03_13
- Smiraglia, R. P. (2009). Bibliocentrism, Cultural Warrant, and the Ethics of Resource Description: A Case Study. *Cataloging & Classification Quarterly*, 47(7), 671–686. <https://doi.org/10.1080/01639370903112013>
- Snow, K. (2017). Defining, Assessing, and Rethinking Quality Cataloging. *Cataloging & Classification Quarterly*, 55(7–8), 438–455. <https://doi.org/10.1080/01639374.2017.1350774>
- Strout, R. F. (1956). The Development of the Catalog and Cataloging Codes. *The Library Quarterly*, 26(4), 254–275. <https://doi.org/10.1086/618341>
- Tennant, R. (2002). MARC must die. *LIBRARY JOURNAL-NEW YORK-*, 127(17), 26–27.
- Tillett, B. (2003). AACR2 and Metadata: Library Opportunities in the Global Semantic Web. *Cataloging & Classification Quarterly*, 36(3–4), 101–119. https://doi-org.utk.idm.oclc.org/10.1300/J104v36n03_09
- Tillett, B. (2013). RDA and the Semantic Web, Linked Data Environment. *JLIS.It : Italian Journal of Library and Information Science*, 4(1), 139–146. <https://doi.org/10.4403/jlis.it-6303>
- U.S. RDA Implementation Updates from the U.S. RDA Test Coordinating Committee. (2016). In *Library of Congress*. U.S. RDA Implementation Updates from the U.S. RDA Test Coordinating Committee. https://www.loc.gov/aba/rda/pdf/RDA_updates_16jan12.pdf
- Virkus, S., & Garoufallou, E. (2020). Data science and its relationship to library and information science: a content analysis. *Data Technologies and Applications*, 54(5), 643–663. <https://doi.org/10.1108/DTA-07-2020-0167>
- Wang, L. (2018). Twinning data science with information science in schools of library and information science. *Journal of Documentation*, 74(6), 1243–1257. <https://doi.org/10.1108/JD-02-2018-0036>

